

Research Document — Domain Understanding & Explainable AI

1. Introduction

1.1 Project scope and Netherlands context

This research addresses **daily bike-rental forecasting** with a Dutch use case in mind. The Netherlands maintains one of the world's highest cycling shares, with official figures indicating that about **28 percent** of journeys are by bicycle, which makes day-to-day demand management operationally important for service quality and costs. Public bike rentals complement rail travel via **OV-fiets**, available at more than 300 rail-linked locations, reinforcing the need to anticipate daily fluctuations rather than react after the fact.

1.2 Motivation and value for daily forecasting

Daily forecasts help operators plan redistribution and maintenance on the right days, for example reducing unnecessary relocations on low-demand weather days, and improving availability when warm and dry conditions are expected. In cities where cycling is integral to first and last mile travel, better day-ahead planning supports user reliability goals and aligns with wider sustainability objectives. Guidance for bikeshare operations repeatedly highlights the importance of anticipating demand to guide rebalancing and service levels.

1.3 Document scope statement

Although the empirical examples later reference the **UCI Bike Sharing Dataset** from Washington, D.C., which provides **daily** rental counts alongside weather and seasonal information, the problem framing is directly relevant to Dutch cities where weather and calendar conditions also drive daily variation. This document establishes the foundation before any modeling, first by clarifying **what Explainable AI means** for a public-mobility forecasting task, second by developing a **domain analysis** that identifies observable **daily** drivers grounded in literature and practice, and third by **deriving data requirements** from the domain rather than from any specific algorithm.

2. Explainable AI, What and Why

2.1 Core definitions and concepts

Explainable Artificial Intelligence refers to AI systems whose outputs can be understood by humans. The literature distinguishes three useful notions, opaque systems that provide predictions without revealing how inputs become outputs, interpretable systems whose internal mapping from inputs to outputs can be analyzed in mathematical terms, and comprehensible systems that accompany outputs with human-readable symbols or descriptions enabling users to form their own explanations. These notions clarify the types of understanding a forecasting tool can provide to planners and riders.

2.2 Why explanations matter in public mobility

Forecasts influence resource allocation, maintenance scheduling, and rider experience, therefore stakeholders need clarity about which observed factors shaped a given daily prediction and why the outcome differs from typical conditions. In European policy, this aligns with Trustworthy AI guidance that emphasizes transparency, human oversight, robustness, privacy and data governance, fairness, societal well-being, and accountability across the AI lifecycle. These rules mean we must clearly explain which factors caused each day's bike-rental prediction in public services.

2.3 Three key principles often guide Explainable AI systems:

1. **Transparency:** The system should clearly show which factors influence its predictions and how those influences work.
2. **Trustworthiness:** Explanations should be consistent, verifiable, and based on real observations, allowing users to assess reliability.
3. **Accountability:** Assumptions, limitations, and decision impacts should be documented so outcomes can be reviewed and improved over time.

These principles are widely recognized in both research and European guidance on **Trustworthy AI**.

3. Domain Analysis

3.1 Research question

Which observable factors influence the daily number of rentals in a city bikeshare system?

This question targets day-to-day variation rather than intraday patterns, because operators plan redistribution and maintenance on a daily cadence, and major public

datasets, including the Washington, D.C. example used in this study, record rentals and weather at a daily level.

3.2 What I do, planned analysis activities

This study synthesizes academic and practitioner evidence to identify measurable daily drivers of bike-rental demand, focusing on weather and calendar conditions that are observable before the operating day. The analysis combines library research on cycling and bikeshare demand with practitioner guidance on system operations, then organizes drivers into weather and calendar categories to inform data requirements for a later modeling phase.

3.3 Why I am doing this, problems and pain points

Operators face two recurring risks, empty stations that harm rider experience during peaks, and idle inventory that increases costs when demand drops. In a cycling-centric context like the Netherlands, even modest percentage shifts in daily cycling volumes can have visible operational consequences, which is why anticipating the influence of temperature, rain, wind, and seasonality is valuable for day-ahead planning. Planning guides emphasize that proactive use of demand expectations improves redistribution efficiency and service reliability.

3.4 Observed factors and their daily effect on the target cnt

Evidence from Dutch and international studies shows consistent daily relationships between weather and cycling activity, higher temperatures and dry conditions are associated with more trips, while precipitation and strong winds correspond to fewer trips. Calendar structure also matters, seasons and months capture broad climatic cycles, while weekdays differentiate commute patterns from weekend leisure riding. These effects provide a defensible basis for constructing daily features that correspond to observable conditions prior to operations.

3.4.1 *Weather factors, temperature, precipitation, wind, humidity*

Daily average or maximum **temperature** is positively associated with cycling up to comfort limits, while **precipitation** reduces usage with stronger effects on heavier rain or snow days. **Wind speed** is typically negatively associated with cycling, and **humidity** can interact with temperature and rainfall in shaping comfort. These variables are readily available from standard meteorological sources at a daily resolution, which aligns with the target definition in public datasets.

3.4.2 *Calendar factors, season, month, weekday, holidays and events*

Season and **month** capture climatic cycles and daylight variation relevant to Dutch conditions, while **weekday** separates commute-oriented demand from weekend leisure patterns. **Holidays and large events** can shift volumes away from typical weekday profiles, so daily holiday indicators and event flags are useful where available.

These calendar variables are deterministic and easy to generate once the analysis period is defined.

3.5 Interim summary of domain insights

At a daily horizon, the most consistent observable drivers are weather and seasonality, specifically temperature, precipitation, wind, and the seasonal or monthly context, complemented by weekday effects and optional holiday or event indicators. These drivers are measurable before operations begin, they are recorded or derivable at daily granularity, and they map directly to decisions that Dutch operators and planners must take for redistribution and maintenance. This evidence-based understanding will be used to define data requirements for a later modeling phase, ensuring that features reflect real-world mechanisms rather than algorithmic convenience.

4. Explainable AI in This Project, Stakeholders and Use Cases

4.1 Company perspective, operations, planning, maintenance

Daily forecasts support company teams in planning bike movements, staff allocation, and maintenance activities without reducing availability. Explanations clarify which observable factors, such as temperature, precipitation, and weekday, influenced the forecast and how it compares to a typical day in similar conditions. These insights help operators improve redistribution planning, shift organization, and maintenance scheduling. Industry guidelines highlight that anticipatory planning becomes more effective when forecasts are transparent and easy to interpret.

4.2 Customer perspective, rider experience and communication

Although this project does not directly communicate with riders, improved operational decisions can enhance customer experience indirectly. When operators use clear and explainable forecasts, bike availability becomes more consistent, and users benefit from a more reliable service. In later applications, forecast insights could also support user-facing communication, such as informing about expected high-demand periods or alternative stations.

4.3 Explanation formats, factor influence summaries, scenario narratives

For operational use, simple explanation formats are most useful. A daily summary can list key factors like temperature, rain, wind, and calendar effects, along with a short note comparing tomorrow's forecast to a normal day. For planners and managers, scenario-based summaries, such as how stronger rain or calmer wind could affect demand, support proactive planning. These formats align with the goal of Explainable AI: helping users understand, trust, and manage AI-assisted decisions appropriately.

4.4 Trustworthy and transparent AI applied to decisions in this context

Because this project provides decision support for operational teams, it applies a minimal set of explanation principles aligned with European Trustworthy AI guidance. These include transparency about which factors shaped each forecast, human oversight when interpreting results, documentation of assumptions and known limits, and fairness checks where relevant. These principles help ensure that explanations remain both technically clear and accountable within the operator's workflow.

5. Data Requirements Based On Domain

5.1 Target definition, daily total rentals

The target variable in this study is the **daily total number of bike rentals** recorded by the system. This approach follows how most public bike-share datasets and operator reports group usage data together with weather and calendar information on a daily basis. Focusing on daily totals keeps the prediction timeline aligned with real operational decisions, such as planning bike redistribution and scheduling maintenance for the next day.

5.2 Input variables, daily weather and calendar features, optional operational signals

Daily weather features such as **temperature**, **rainfall**, **wind speed**, and **humidity** are key factors that affect cycling activity and can be taken from national weather services. Calendar features like **season**, **month**, and weekday capture climate cycles and travel habits, while holiday or event flags explain unusual days. Optional operational data, such as available bikes, docks, or maintenance times, can help separate real demand changes from supply limits when available..

5.3 Granularity and time span, daily aggregation and seasonal coverage

All inputs should be **aggregated or sourced at daily granularity** to match the target. The training period should cover **at least one full annual cycle** to expose the model and explanations to seasonal variation. If possible, include two years to improve robustness of seasonality estimates and to support simple out-of-time validation at the daily level.

5.4 Data quality and governance, units, missing values

Define **units and encoding** up front, for example temperature in degrees Celsius, precipitation in millimeters per day, wind speed in meters per second, and calendar variables as categorical features. Specify a **missing-value policy** for weather and target fields, for example drop days with missing target.

5.5 Candidate data sources, to be confirmed later

- **Bike-share usage**, the UCI **Bike Sharing** dataset from Washington, D.C., provides daily rental counts together with weather and seasonal information and serves as a well documented public exemplar for field names and daily structure during the Design phase.
- **Dutch weather**, the **KNMI** open data platform exposes national observations and documentation for station data and daily variables that can supply temperature, precipitation, wind, and humidity at the required resolution.
- **Service context**, operator and planning materials illustrate how daily forecasts inform redistribution and service levels, these sources help align variable choices and explanation formats with real operational decisions in the Netherlands and abroad.

6. Research Methodology Note, DOT Framework

In this study, I used the Development Oriented Triangulation (DOT) Framework to guide my research through three strategies. I conducted **library research** to review academic and practitioner sources on bike-rental forecasting and Explainable AI, ensuring a strong theoretical foundation. I carried out a **community scan** to gather insights from developer and operator discussions, helping identify real operational challenges and practical variables. Finally, I used **showroom validation** by comparing my findings with public reports from Dutch operators to confirm their relevance and alignment with real-world practices. This combination ensured that my work was both evidence-based and grounded in practice.

7. References and Resources

1. Doran, D., Schulz, S., & Besold, T. R. (2017). *What Does Explainable AI Really Mean? A New Conceptualization of Perspectives*. arXiv preprint 1710.00794. **arXiv:** <https://arxiv.org/abs/1710.00794>
2. European Commission High-Level Expert Group on AI. (2019). *Ethics Guidelines for Trustworthy AI*. **Publications Office of the EU:** <https://op.europa.eu>
3. DARPA. *Explainable Artificial Intelligence (XAI) Program*. **AI Magazine / AAAI OJS (retrospective):** <https://ojs.aaai.org>
4. UCI Machine Learning Repository. *Bike Sharing Dataset* (Capital Bikeshare, Washington, D.C., daily and hourly counts with weather and seasonal information). **UCI:** <https://archive.ics.uci.edu/ml/datasets/bike+sharing+dataset>

5. KNMI — Royal Netherlands Meteorological Institute. *KNMI Data Platform* (daily weather variables for NL). **KNMI:** <https://dataplatform.knmi.nl>
6. ITDP. *The Bikeshare Planning Guide*. **ITDP:** <https://www.itdp.org>
7. ITDP. (2018 and updates). *The Bikeshare Planning Guide* (planning, redistribution, service levels). itdp.org+2TUMI+2
8. Nederlandse Spoorwegen (NS). *OV-fiets* overview and rental locations. **NS:** <https://www.ns.nl/ov-fiets>
9. KiM Netherlands Institute for Transport Policy Analysis. *Cycling facts* (English portal). **KiM:** <https://english.kimnet.nl>
10. Sport en Bewegen in Cijfers. *Cycling in the Netherlands* facts and datasets. **SBC:** <https://www.sportenbewegenincijfers.nl>
11. European Commission. *European approach to AI* and AI Act policy pages. **Digital Strategy:** <https://digital-strategy.ec.europa.eu>
12. ScienceDirect — transport and health journals used for weather–cycling studies. **ScienceDirect:** <https://www.sciencedirect.com>
13. SpringerLink — transportation research used for Dutch cycling demand studies. **SpringerLink:** <https://link.springer.com>